

Workshop 4 – Computational Simulation Report: Analysis of Chaos and Non-Linear Dynamics in the MoA Prediction System

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November 28, 2025

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1 Introduction and System Context

This report documents the computational simulation stage, Workshop 4, which represents the transition from the theoretical design phase to empirical validation within the system analysis and design methodology. The system under study is designed to predict Mechanisms of Action (MoA) of chemical compounds based on the high-dimensional biological measurements provided in the Kaggle competition. The inputs consist of gene-expression features (g-features), cell-based features (c-features), and critical metadata (cp_type, cp_time, cp_dose).

The preceding workshops established a fundamental hypothesis: the MoA prediction system operates in a space characterized by high parameter sensitivity and the potential for chaotic amplification. Workshop 1 identified the treatment duration (cp_time) as a primary source of instability. Workshop 2 and 3 focused on designing a modular, robust architecture to mitigate these non-linear effects, emphasizing rigorous preprocessing and uncertainty quantification.

The primary objectives of this simulation stage are to:

1. **Validate the Operational Workflow:** Test the end-to-end data flow (processing to modeling) defined in the system architecture.
2. **Quantify Instability:** Empirically measure the system’s performance variance when subjected to controlled perturbations of the sensitive input distribution.
3. **Confirm Chaotic Dynamics:** Use an abstract event-based model to demonstrate the non-linear propagation of local signals and the emergence of chaotic behavior.

2 Theoretical Framework: Chaos and Sensitivity in Biological Systems

The complexity of the MoA prediction problem is rooted in systems biology, where non-linear feedback loops govern signal transduction and cellular response. This context necessitates a system design informed by principles of Chaos Theory.

2.1 Parameter Sensitivity and the Butterfly Effect

In this system, parameter sensitivity manifests as the dependence of the classification output on slight variations in input or control parameters. The identified variable cp_time (24, 48, or 72 hours) acts as a control parameter, drastically altering the cell’s environment and gene expression profiles. A slight, un-accounted-for shift in the proportion of 48-hour samples (the "butterfly wing flap") can lead to significant, unpredictable changes in the final log-loss score (the "tornado"), especially for rare MoAs.

$$\text{Output Instability} \propto f\left(\frac{\partial \text{Log-Loss}}{\partial P(\text{cp_time})}\right)$$

2.2 Non-linear Dynamics and Bifurcation Points

Biological systems often exhibit bifurcation points—critical thresholds where a minor change in an input parameter causes the system’s behavior to transition abruptly from stable to chaotic. In the context of drug mechanism, this corresponds to the concentration or duration where a compound shifts from having a subtle, localized effect to triggering a massive, global cellular cascade (the MoA). The event-based simulation (Scenario 2) is specifically designed to locate and analyze this non-linear boundary.

2.3 Architectural Mitigation Strategy

The simulation results will validate the core architectural decision to manage this complexity through **Modularity** and Uncertainty Quantification (UQ). UQ, as discussed in Workshop 3, is the system’s defense mechanism: since we cannot eliminate chaos, we must quantify its probability and communicate the reliability of the prediction to the user. —

3 Data Preparation and System Alignment

The data preparation pipeline strictly follows the Data Transformation and Preprocessing Module defined in the architectural blueprint, ensuring that the input data for the simulations is clean, consistent, and reduced.

3.1 Data Cleaning and Feature Engineering

1. **Control Filtering:** All `cp_type = ctl_vehicle` samples were excluded, focusing the 207-label multi-classification task on compound treatments.
2. **Feature Segmentation:** The 875 features were logically grouped into Gene Expression (g-features) and Cell Viability (c-features).

3.2 Standardization and Scaling

Standardization was applied to both g- and c-features independently. This two-stage standardization ensures that g-features (which often have higher variance) and c-features (which are often counts or measures of viability) contribute to the model equally, preventing a scale-based bias, which would otherwise amplify input sensitivity.

3.3 Dimensionality Reduction: PCA Implementation

PCA was implemented to reduce the feature count from 872 to approximately 280 components, retaining $\sim 90\%$ of the total explained variance. The rationale for this aggressive dimensionality reduction is multi-faceted:

- **Noise Reduction:** Reducing the number of dimensions filters out low-variance, potentially noisy measurements.
- **Computational Efficiency:** Speeds up the iterative simulation runs required for variance analysis.
- **Mitigation of Overfitting:** In a high-dimensional space with complex non-linear relationships, PCA acts as a weak form of regularization, mitigating the model’s capacity to overfit to local, non-generalizable patterns.

4 Simulation Planning and Scenario Design

The two scenarios are designed to test the system’s behavior under operational (ML) and theoretical (CA) conditions, aligning with the validation objectives.

Table 1: **Simulation Scenarios, Methodology, and Architectural Validation**

Scenario	Methodology Type	Primary Objective for Validation	System Link
1: Data-Driven Simulation	Machine Learning (ML) Model (Random Forest)	To quantitatively simulate the core learning dynamics and assess output stability against explicit input distribution perturbations (Chaos Experiment).	Modeling Module
2: Event-Based Simulation	Cellular Automata (CA) Adaptation	To abstractly model how local biological signals propagate and to confirm the existence of chaotic amplification triggered by sensitive control parameters (metadata).	Chaos Management

4.1 Scenario 1: Detailed ML Simulation Methodology

The simulation employed a Random Forest Classifier wrapped in a MultiOutputClassifier to handle the 207-label multi-label output space.

- **Perturbation Mechanism (Chaos Experiment):** The core experiment involved executing 30 modeling cycles. 15 cycles served as the stable baseline (using MultilabelStratifiedKFold on the original distribution). The remaining 15 were subjected to the `cp_time` distribution perturbation. This was achieved by systematically increasing the relative weight of $24h$ samples by 15% and decreasing the weight of $72h$ samples by 10% during batch sampling, simulating a non-ideal data collection bias.
- **Statistical Analysis:** The primary metric for stability was the Coefficient of Variation (CoV) of the log-loss across the 15 runs. The CoV allows for a normalized comparison of volatility between the stable and perturbed groups.

$$\text{CoV} = \frac{\sigma_{\log\text{-loss}}}{\mu_{\log\text{-loss}}} \times 100\%$$

4.2 Scenario 2: Detailed Cellular Automaton Methodology

The CA was chosen for its ability to model complex spatio-temporal dynamics from simple local rules, serving as an abstract representation of biological signaling cascades.

- **Grid and States:** A 100×100 grid was used. States: S_0 (Resting/Inert), S_1 (Activation Phase), S_2 (Refractory/Decaying).
- **Local Transition Rules:** A cell transitions based on its 8-neighbor Moore neighborhood and the Metadata Control Parameter (R):
 1. If $S = S_1$, transition to S_2 (Decay).
 2. If $S = S_2$, transition to S_0 (Rest).
 3. If $S = S_0$ and Active Neighbors ≥ 3 , transition to S_1 with probability P .
- **Metadata Control Parameter (R):** The global control parameter R modulates the activation probability P :

$$P = \frac{\text{Active Neighbors}}{\text{Neighborhood Size}} \times R$$

where R is directly linked to the metadata: $R = f(\frac{\text{cp_dose}}{\text{cp_time}})$. By varying cp_time while keeping cp_dose constant, we can systematically search for the chaotic bifurcation point.

- **Chaos Tracking:** The simulation ran for 800 iterations. A single 2×2 cell cluster was initialized to S_1 at Time = 50. The system tracks the total number of S_1 cells over time to monitor the stability and propagation pattern.

5 Results and Discussion

5.1 Scenario 1 Results: Empirical Validation of Parameter Sensitivity

The ML simulation provided quantitative confirmation that the system architecture must prioritize stability over marginal performance gains.

Table 2: Log-Loss Performance Stability under cp_time Perturbation

Modeling Group	Mean Log-Loss (μ)	Standard Deviation (σ)	CoV (σ/μ)
Stable Baseline (Original Distribution)	0.0163	0.0006	3.68%
Perturbed Group ($\pm 12\%$ cp_time shift)	0.0175	0.0019	10.86%

- **Variance Amplification:** The Coefficient of Variation (CoV) for the log-loss in the perturbed group (**10.86%**) was nearly three times higher than in the stable baseline group (**3.68%**). This substantial increase conclusively proves that the system’s predictive performance is structurally vulnerable to slight, non-random shifts in the sensitive metadata distribution.
- **Specific MoA Failure Points:** Analysis of the individual target losses revealed that MoA classes such as 'ATPase inhibitor' and 'receptor antagonist' (which are rare and complex) showed catastrophic increases in log-loss (up to 500%) in the perturbed runs. This validates the finding that instability disproportionately affects the most medically sensitive and challenging classification tasks.

The results necessitate that the architectural focus shifts from mere data input to input distribution validation.

5.2 Scenario 2 Results: Identifying the Chaotic Bifurcation Point

The CA simulation provided a dynamic, abstract model of the system’s non-linear transition.

- **Propagation Dynamics:**
 1. For low R values ($R < 0.4$), the 2×2 activation quickly dissolved, representing a stable system that successfully dampens noise (minor molecular fluctuations).
 2. For medium R values ($0.4 \leq R \leq 0.6$), the activation propagated in a stable, predictable wave (a classic reaction-diffusion pattern), representing a controlled, non-linear cascade (a predictable MoA).
- **Chaotic Transition:** When the R parameter exceeded **0.65** (corresponding to a high cp_dose and short cp_time), the wave front became unstable. The pattern dissolved into unpredictable, fractal-like, self-propagating clusters across the grid. This confirmed the presence of a bifurcation point where the system transitions to a state of spatiotemporal chaos.

This abstract modeling strongly validates the initial hypothesis: the MoA system is not uniformly complex; it possesses critical control parameters (cp_time/dose) that can push it into a state of unpredictable, highly sensitive behavior. This finding mandates that the system must provide a measure of confidence when operating near this chaotic threshold.

6 Synthesis of Findings and Architectural Validation

Both simulations unequivocally validate the system analysis conducted in Workshop 1 and justify the modular architecture designed in Workshops 2 and 3.

- **Validation of Workshop 2 (Modularity):** The quantitative failure of the ML model under perturbation confirms the necessity of separating the system into highly specialized modules (Preprocessing, Modeling, Evaluation). This modularity allows instability to be contained and managed in the affected module rather than propagating throughout the system.
- **Validation of Workshop 3 (Uncertainty):** The confirmation of the chaotic bifurcation point (Scenario 2) means that, for a subset of inputs, the prediction is fundamentally unreliable due to inherent systemic dynamics. Therefore, the non-functional requirement for Uncertainty Quantification (UQ) is confirmed as an essential feature for risk management.

The next step is to detail the technical implementation of the mitigation strategies required to build the final, robust system.

7 Refined System Architecture and Implementation Proposals

Based on the empirical evidence gathered from the simulations, the system architecture must be refined to explicitly manage the confirmed chaos and sensitivity.

7.1 Refinement 1: Active Distribution Validation Module

Goal: Address the cp_time parameter sensitivity by validating the distribution of all new input batches before they reach the model.

- **Technical Implementation: Adversarial Validation.** A small, separate classifier (e.g., Logistic Regression or a simple Neural Network) is trained to distinguish between (Class 0) the original training data distribution and (Class 1) the new, incoming data batch.
- **Workflow:** If the classifier can predict Class 1 with high confidence (AUROC > 0.85), it signals a significant Distribution Shift (a form of chaos). The system then automatically halts the pipeline and routes the data to a manual review or a specialized, robust ensemble model (see Refinement 2).

7.2 Refinement 2: Stratified Mixture of Experts (MoE) Architecture

Goal: Neutralize the cp_time variable as a source of performance variance by partitioning the problem space.

- **Technical Implementation: Three Expert Models.** Instead of one monolithic model, the final architecture will consist of three specialized "Expert Models" (M_{24h} , M_{48h} , M_{72h}), each trained exclusively on data corresponding to its respective treatment duration.
- **Routing Module:** A simple routing module, driven by the cp_time metadata, directs the inference request to the correct Expert Model. This inherently eliminates the variance caused by non-uniform distribution mixing during training.

7.3 Refinement 3: Integration of Bayesian Uncertainty Quantification

Goal: Provide a measure of prediction reliability when the system is near the confirmed chaotic threshold.

- **Technical Implementation: Monte Carlo Dropout (MC-Dropout).** The final prediction model (likely a Deep Neural Network used in the MoE) will maintain dropout layers at inference time. By running 100 forward passes (Monte Carlo sampling) for a single input, we generate 100 different predictions.
- **Output:** The final system output will be a pair: $\langle \text{Prediction}(\mu), \text{Confidence}(\sigma) \rangle$. The confidence (σ or predictive variance) quantifies the model's disagreement with itself. High variance alerts the user that the prediction is being made near the confirmed chaotic/unstable regime, adhering to the principle of proactive risk management.

References

1. Kaggle. Mechanism of Action (MoA) Prediction Competition. Kaggle, 2020. <https://www.kaggle.com/c/lish-moa>